

CROSS-LINGUAL ACTIVATION STEERING IN LARGE
LANGUAGE MODELS: REPRESENTATIONAL ALIGNMENT AND
BEHAVIOURAL TRANSFER

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ABSTRACT

Large language model alignment methods are often developed and evaluated in English, but their behaviour may not transfer consistently across languages. This dissertation investigates the cross-lingual generalisation of Contrastive Activation Addition (CAA) for reducing sycophantic behaviour in large language models. We construct and apply steering directions across three open-weight models and five languages: English, Italian, Chinese, Korean, and Kazakh. We evaluate the interventions using held-out test sets, layer-wise analysis, bootstrap confidence intervals, and random and permutation controls. We additionally construct separate English-derived truthfulness steering vectors for a forced-choice TruthfulQA task to examine whether cross-lingual transfer is behaviour-dependent.

Our experiments show that cross-lingual transfer is model- and behaviour-dependent. English-derived sycophancy steering transferred reliably to all four non-English languages in Qwen3.5-9B and Gemma4-12B-IT, but not in OLMo3-7B-Instruct, where only the English effect was statistically reliable. A cross-task evaluation revealed a dissociation: Qwen3.5-9B, which transferred sycophancy steering reliably, showed no consistent pattern of cross-lingual truthfulness transfer at the English-selected layer, whereas OLMo3-7B-Instruct, which failed to transfer sycophancy, produced reliable truthfulness gains in three of four non-English languages.

Across models and behaviours, strongly aligned steering directions produced uneven behavioural transfer, and in OLMo3-7B-Instruct no non-English sycophancy effect was statistically reliable. High cross-lingual representational similarity is therefore not sufficient to guarantee comparable behavioural effects. Whether such similarity is necessary remains unresolved. These results indicate that the cross-lingual reliability of alignment interventions should be evaluated through behavioural outcomes in each target language, rather than inferred from representation geometry alone.

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INTRODUCTION

Motivation. Large language models (LLMs) are increasingly used across languages, while many alignment methods are still developed and evaluated primarily in English. An intervention that changes model behaviour successfully in English may therefore not produce the same effect when the model is used in another language. This is particularly important for behavioural alignment methods, where the goal is not only to modify an internal representation but to produce a reliable change in model outputs.

Activation steering provides a useful setting for studying this problem. Rather than updating model parameters, activation steering modifies intermediate representations during inference using directions associated with particular behaviours. Contrastive Activation Addition (CAA) constructs these directions from differences between contrasting examples and has been used to steer behaviours such as truthfulness, harmlessness, and sycophancy [1]. Because multilingual LLMs often share substantial representational structure across languages, a steering direction learned in one language may also transfer to others. However, similarity between internal directions does not necessarily mean that the resulting behavioural effect will be equally strong.

This dissertation focuses on sycophancy, where a model tends to agree with a user’s stated view rather than provide an independent answer [2]. Sycophancy provides a useful test case because the target behaviour can be measured directly before and after an intervention. It also allows the same behavioural contrast to be evaluated across multiple languages while keeping the intervention procedure fixed.

Objective. The aim of this dissertation is to examine how reliably CAA steering transfers across languages and behaviours, and whether representational similarity is a good indicator of behavioural transfer. We study three open-weight language models and five languages: English, Italian, Chinese, Korean, and Kazakh. English-derived steering directions are applied across languages, and their effects are measured over multiple layers using held-out evaluation sets.

The experiments examine the problem from both behavioural and representational perspectives. At the behavioural level, we measure changes in sycophancy after steering and compare the strength of the intervention across languages. At the representational level, we compare language-specific steering directions using cosine similarity and vector norms. Bootstrap confidence intervals are used to estimate uncertainty in the measured effects, while random and permutation

controls test whether similar changes can be produced by arbitrary perturbations. We also compare English-derived steering with directions constructed directly in the target languages.

The main question is whether a steering direction that appears shared in representation space also produces a comparable behavioural intervention across languages. The results show that this correspondence is incomplete: steering directions can remain highly similar across languages while producing different behavioural effects, and transfer patterns can differ across behaviours within the same model.

Research Questions. This dissertation addresses the following research questions:

1. To what extent do CAA steering directions learned in English reduce sycophantic behaviour across different languages, and does this pattern hold across model families?
2. How similar are steering directions across languages, and to what extent does this representational similarity correspond to behavioural transfer?

Contributions. The main contributions of this dissertation are three-fold. First, we provide a controlled cross-lingual evaluation of activation steering across three open-weight LLMs and five languages with different typological and resource characteristics, including a cross-task evaluation on TruthfulQA to examine whether transfer patterns generalise across behaviours. Second, we compare behavioural transfer with the geometry of language-specific steering directions, allowing representational similarity and intervention effectiveness to be evaluated separately. Third, we use held-out evaluation, bootstrap confidence intervals, and random and permutation controls to test whether the observed cross-lingual effects are robust to sampling variation and arbitrary activation perturbations.

Together, these experiments provide evidence that successful multilingual transfer cannot be inferred from representational similarity alone, and that a model’s cross-lingual transfer profile can differ substantially across behaviours. Cross-lingual steering therefore needs to be evaluated directly in terms of the behaviour produced in each target language.

LITERATURE REVIEW

2.1 LLMS AND ALIGNMENT

Large language models (LLMs) have developed from task-specific systems into general-purpose models capable of adapting to a wide range of tasks through natural-language context. Brown et al. [3] demonstrated that increasing model scale enables few-shot generalisation across tasks including translation, question answering, and reasoning without task-specific parameter updates. This ability allows a single model to respond flexibly to new instructions and user-provided information.

However, sensitivity to user context does not necessarily imply alignment with the user's underlying interests or with standards of factual and independent reasoning. A model may adapt appropriately to information supplied by the user, but it may also adjust its answer merely to agree with the user's stated beliefs or preferences. This latter behaviour is commonly described as sycophancy. Sycophantic responses can reinforce incorrect beliefs, reduce the reliability of model outputs, and make model agreement difficult to distinguish from independently supported judgement.

Perez et al. [2] introduced model-written evaluations as a scalable method for identifying and measuring undesirable model behaviours, including sycophancy. Their evaluation items pair a user profile or expressed opinion with alternative responses that either match or contradict the user's position. This controlled structure makes it possible to measure whether a model systematically favours user-aligned answers, and it later became a useful basis for evaluating activation-level interventions targeting sycophancy.

A prominent approach to language-model alignment is Reinforcement Learning from Human Feedback (RLHF). Bai et al. [4] developed the Helpful, Honest, and Harmless framework, using human preference judgements to optimise model responses toward desired behavioural properties. Preference-based optimisation can substantially improve the perceived helpfulness and safety of model outputs. Nevertheless, optimising for preferred responses does not by itself establish whether a model is responding independently or merely producing answers that appear agreeable to users.

Preference optimisation also primarily changes model behaviour through an external training signal. It does not directly identify how a behaviour such as sycophancy is represented within the model or whether that representation can be manipulated at inference time.

Addressing these questions requires methods that examine and intervene on internal activations. Activation steering provides one such approach by extracting candidate behavioural directions from model representations and testing whether modifying those directions produces a corresponding change in output behaviour.

2.2 ACTIVATION STEERING

Activation steering refers to a family of inference-time methods that modify a model’s internal activations to influence its outputs without updating model parameters. Unlike fine-tuning or reinforcement learning, these methods intervene directly on selected hidden-state directions during generation. This makes activation steering useful not only as a lightweight behavioural intervention, but also as a tool for testing whether a hypothesised internal representation has a causal effect on model output.

Turner et al. [5] introduced Activation Addition (ActAdd), showing that vectors derived from activation differences between contrastive prompts could induce targeted changes in model generation. Adding or subtracting a concept-related direction produced corresponding changes in output content, providing early evidence that semantically meaningful information can be represented approximately as directions in the residual stream. However, ActAdd relied on a small number of manually selected prompt pairs, which may limit the stability and generality of the extracted directions.

Rimsky et al. [1] subsequently proposed Contrastive Activation Addition (CAA), which forms the methodological basis of this dissertation. CAA averages activation differences across a dataset of contrastive examples representing opposing behavioural responses, producing one steering vector per layer. The vector is then added to the residual stream with a scalar multiplier controlling the strength and direction of the intervention. CAA was evaluated on behaviours including sycophancy, corrigibility, and self-preservation, demonstrating that dataset-level contrastive activation differences can capture behaviourally relevant directions. Its use of structured multiple-choice evaluations also enables intervention effects to be measured quantitatively.

Related methods differ in how steering directions are extracted and where interventions are applied. Representation Engineering (RepE; Zou et al. 6) uses techniques such as principal component analysis to identify high-level representational directions, while Inference-Time Intervention (ITI; Li et al. 7) modifies selected attention-head outputs rather than the full residual stream at a single layer. These approaches demonstrate that both the extraction method and intervention location can influence steering effectiveness.

Activation steering is nevertheless not uniformly reliable. Tan et al. [8] documented substantial variation in per-example steerability and identified anti-steerability, in which an intervention changes behaviour in the opposite direction from that intended. These findings show that the existence of an extracted direction does not guarantee a consistent behavioural effect across examples or layers. Steering methods must therefore be evaluated empirically using layer-wise analysis and appropriate control conditions.

Recent work has also explored adaptive alternatives to fixed-vector intervention. Ye et al. [9], for example, proposed RISER, which selects and composes reusable reasoning vectors according to the input. Although adaptive methods may provide greater flexibility, they introduce additional routing and optimisation components. The present study instead uses fixed CAA vectors in order to isolate whether a behavioural direction extracted from English transfers directly across languages without language-specific adaptation.

Most activation-steering research has focused on English or monolingual settings. Maraia et al. [10] directly examined whether activation steering constructed from English syllogistic reasoning examples generalises across languages. They found substantial multilingual improvements in formal reasoning, providing evidence that English-derived steering can transfer beyond the source language. Their setting, however, targets reasoning performance rather than alignment behaviours such as sycophancy. It therefore remains unclear whether behavioural directions extracted in one language produce comparable effects across typologically diverse languages, whether such transfer depends on language-specific representational structure, and whether transfer reliability corresponds to cross-lingual representation geometry. Addressing these questions requires examining both behavioural transfer and the geometric relationships among independently extracted language-specific vectors.

2.3 REPRESENTATION GEOMETRY

Activation steering is motivated by the possibility that behaviourally relevant information is organised in a form that can be recovered and manipulated within a model’s activation space. If no such structure were present, injecting a vector into the residual stream would be expected to produce nonspecific disruption rather than a coherent behavioural change.

Park et al. [11] formalised this intuition through the Linear Representation Hypothesis. Their framework considers the conditions under which high-level concepts may be represented as linear directions and distinguishes ordinary Euclidean geometry from a causal inner product that reflects how representational changes affect downstream computation. Under this account, standard geometric measures can

describe relationships between vectors without establishing that the vectors have equivalent behavioural effects.

Two language-specific vectors may exhibit high cosine similarity while interacting differently with downstream computations. A direction may therefore appear well aligned under a conventional geometric measure but produce a different intervention effect when applied in another linguistic context. Geometric alignment should consequently be interpreted as a descriptive or explanatory signal rather than as direct evidence of causal equivalence.

Marks and Tegmark [12] provided empirical evidence that truth-related information can exhibit recoverable linear structure in language-model representations. Their results showed that factual truth values can be separated using linear methods, supporting the broader possibility that high-level semantic properties may be organised directionally in activation space. Although their analysis concerns truthfulness, it motivates examining whether other alignment-relevant behaviours, including sycophancy, exhibit similarly recoverable structure.

The layer at which a vector is extracted may also affect its stability and behavioural relevance. Rimsky et al. [1] reported greater directional stability in middle-to-later portions of the network than in earlier layers. This pattern is consistent with the possibility that behaviourally relevant information becomes more clearly organised at greater depth, although layer function should not be reduced to a simple lexical-to-semantic hierarchy. The observed layer dependence nevertheless provides a rationale for screening intervention effects across the network before selecting model-specific layer ranges.

These findings suggest that three related properties should be distinguished: whether a behavioural direction can be recovered, whether it is geometrically stable across layers or languages, and whether intervention along that direction produces the intended behavioural effect. In multilingual settings, these properties may diverge. Language-specific vectors may be directionally similar while differing in magnitude or behavioural effectiveness, or they may be geometrically distinct despite supporting comparable transfer effects.

The present study therefore compares independently extracted language-specific vectors using pairwise cosine similarity and vector norms. These measures are used to examine whether representational geometry helps explain variation in cross-lingual steering effectiveness, while avoiding the stronger claim that geometric similarity alone establishes causal transfer.

2.4 CROSS-LINGUAL TRANSFER

Multilingual language models process multiple languages within a shared parameter space, creating the possibility that some semantic and behavioural information is represented in partially common form

across languages. However, differences in tokenisation, morphology, writing system, training-data coverage, and language-specific processing may also produce distinct activation patterns. A behavioural direction extracted from English may therefore interact differently with representations elicited by Korean, Kazakh, or other target-language inputs.

Zhao et al. [13] show that language-specific and reasoning-related representations in multilingual LLMs can be partially disentangled, with causal interventions on language-specific representations affecting multilingual reasoning performance. Their findings suggest that language-related internal representations can influence task-level behaviour rather than acting only as surface-level encodings. For activation steering, this means that successful intervention in English does not guarantee an equivalent effect in another language.

Cross-lingual divergence may not correspond directly to typological distance. A language that is structurally distant from English may still activate a compatible behavioural representation, while a more closely related language may differ at the layer where the intervention is applied. Transfer may therefore depend less on surface similarity alone than on whether the target-language activations support a direction that is functionally compatible with the English-derived steering vector.

Kirtane and Huang [14] derived language vectors from activation differences between source- and target-language examples and used them to steer multilingual in-context learning. Their results show that language-related directions exhibit meaningful linguistic structure and can transfer across tasks, supporting the view that language identity has structured directional organisation within activation space. This raises the possibility that language-specific and behaviour-specific information coexist within model representations and may interact during steering.

Pokharel et al. [15] proposed Cross-Lingual Activation Steering (CLAS), a training-free method that modulates neuron activations to improve performance in non-dominant languages. Their findings indicate that successful multilingual intervention does not necessarily require target-language representations to align as closely as possible with English. Instead, intervention effectiveness may depend on the functional organisation of shared and language-specific representations.

CLAS differs from the present study in both its intervention mechanism and research objective. It selectively modulates neuron-level activation patterns to improve multilingual task performance, whereas this dissertation evaluates whether a fixed sycophancy-related direction extracted through English CAA produces behavioural effects when applied without modification to other languages. The present

study also includes Korean and Kazakh, for which evidence concerning English-derived sycophancy steering remains limited.

Related work on cross-model steering further shows that activation interventions can be transferred across representational contexts when an appropriate mapping is learned. Oozeer et al. [16] demonstrated that steering interventions can be transferred between model activation spaces using learned mappings, while direct or simpler mappings may perform less reliably across substantially different architectures. Although cross-model and cross-lingual transfer are distinct settings, this work illustrates that conceptual correspondence does not necessarily imply direct compatibility between representational spaces.

The present study does not learn language-specific mappings or tune the intervention on target-language results. Instead, the same English-derived vector is applied directly to parallel inputs in each target language. This design evaluates zero-shot transfer across linguistic realisations while controlling the underlying item content. Independently extracted language-specific vectors are then compared to examine whether variation in behavioural transfer is associated with directional or magnitude-based differences in representation geometry.

2.5 RESEARCH GAP

Prior work establishes that activation steering can modify model behaviour through inference-time intervention and that behaviourally relevant information may exhibit recoverable directional structure within model activations. Emerging multilingual studies further indicate that activation-level interventions do not transfer uniformly across languages and that geometric alignment does not necessarily imply equivalent behavioural effects.

Nevertheless, it remains unclear whether a sycophancy-related steering vector extracted from English can produce consistent behavioural changes across typologically diverse languages without language-specific adaptation. Existing evidence is particularly limited for Korean and Kazakh, whose morphology, writing systems, and training-data representation differ substantially from English. Evaluating these languages alongside Italian and Chinese provides a broader test of cross-lingual steering under varied linguistic conditions.

A second unresolved question concerns the relationship between behavioural transfer and representation geometry. Independently extracted language-specific vectors may differ in direction, magnitude, or layer-wise organisation. It is not yet established whether greater cosine similarity to the English vector corresponds to stronger cross-lingual steering, or whether comparable behavioural effects can emerge despite geometric divergence. Cosine similarity and vector norms

must therefore be interpreted as complementary geometric measures rather than as direct indicators of behavioural transfer.

A third methodological issue is whether observed behavioural changes depend on the learned sycophancy direction or merely reflect non-specific perturbation of the residual stream. Norm-matched Gaussian and permuted vectors provide controls for testing whether the learned direction produces effects beyond those caused by perturbations with comparable magnitude or component distributions. Evaluating multiple intervention layers across three model families further examines whether the observed effects are confined to a particular representational region or model family.

This dissertation addresses these gaps by extracting English CAA vectors from sycophancy data and applying them without modification to English, Italian, Chinese, Korean, and Kazakh inputs. English is used as an in-distribution source-language reference, while the target-language conditions evaluate zero-shot transfer across parallel linguistic realisations. The study compares learned interventions with norm-matched controls and separately extracts language-specific vectors for geometric analysis using pairwise cosine similarity and vector norms.

The dissertation therefore investigates two related questions: whether an English-derived sycophancy direction produces transferable behavioural effects across languages, and whether variation in those effects is associated with the geometry of independently extracted language-specific directions. Cross-task evaluation on TruthfulQA further examines whether transfer patterns are consistent across behaviours, or whether the same model may show different profiles on different tasks.

METHODOLOGY

3.1 CONTRASTIVE ACTIVATION ADDITION

Contrastive Activation Addition [CAA; 1] is used as the primary steering method in this study. CAA enables inference-time intervention without updating model parameters and produces explicit steering vectors that can be compared across languages using directional and magnitude-based analyses. It is therefore well suited to evaluating both the behavioural transfer and representational geometry of sycophancy-related directions.

Two distinct vector-extraction settings are used in this study. For the behavioural transfer experiments, steering vectors are extracted exclusively from English data and applied to target-language inputs without modification. For the representational geometry analysis, additional language-specific vectors are extracted independently from the corresponding translated datasets using the same procedure. These language-specific vectors are used only for geometric comparison and are not applied in the zero-shot behavioural transfer experiments.

3.1.1 Steering Vector Extraction

CAA constructs a steering vector from contrastive pairs that differ in sycophantic versus non-sycophantic behaviour. Given a dataset of N contrastive pairs $(x_i^+, x_i^-)_{i=1}^N$, where x_i^+ represents a sycophantic response and x_i^- represents a non-sycophantic response, the steering vector at layer ℓ is computed as:

$$v_\ell = \frac{1}{N} \sum_{i=1}^N (h_\ell(x_i^+) - h_\ell(x_i^-)) \quad (1)$$

where $h_\ell(x)$ denotes the residual stream activation at layer ℓ for input x , extracted at the token position immediately preceding the answer choice, following a multiple-choice activation extraction setup adapted from Rimsky et al. [1].

For the behavioural transfer experiments, the vector is extracted from English data and applied to target-language inputs without modification. For the representational geometry analysis, the same extraction procedure is applied separately to each language-specific dataset, producing one vector per language and layer.

3.1.2 Activation Intervention

At inference time, the steering vector is added to the residual stream at every forward pass through the target layer:

$$h_\ell^{\text{steered}} = h_\ell + \alpha v_\ell \quad (2)$$

where α is a scalar multiplier controlling the strength and direction of the intervention. A negative multiplier, $\alpha < 0$, steers the model away from the sycophantic direction, while a positive multiplier amplifies it.

3.1.3 Direction and Multiplier Choice

Negative multipliers are used to steer the models away from the sycophantic direction represented by the extracted vector. Intervention strength is selected separately for each model through preliminary multiplier sweeps on English data.

For Qwen3.5-9B and OLMo3-7B-Instruct, we use $\alpha = -1.5$, which produced a clear reduction in sycophancy while preserving a high proportion of parseable outputs. For Gemma4-12B-IT, we use $\alpha = -1.0$. A stronger intervention of $\alpha = -1.5$ caused substantial degradation in output quality, including an increase in refusal-like and unparseable responses, and was therefore not retained.

Intervention layers are also selected separately for each model through preliminary full-layer sweeps on English data. Subsequent experiments focus on layers 13–18 for Qwen3.5-9B, layers 24–28 for Gemma4-12B-IT, and layers 12–16 for OLMo3-7B-Instruct, with L14 serving as the English-selected reference layer for OLMo3. Target-language results are not used to select either the multiplier or intervention layers.

3.2 MODELS

We evaluate the steering procedure on three open-weight language models of broadly similar parameter scale but different internal architectures: Qwen3.5-9B, Gemma4-12B-IT, and OLMo3-7B-Instruct.

3.2.1 Qwen3.5-9B

Qwen3.5-9B [17] is a 9-billion-parameter open-weight model with a hybrid architecture combining linear-attention and full-attention layers.

The model checkpoint used in this study is Qwen/Qwen3.5-9B.

The configuration comprises 32 layers with a hidden dimension of 4096. Qwen3.5 supports multilingual generation and uses a thinking inference mode by default. This mode is disabled in all experiments

to obtain consistent and directly parseable outputs, as detailed in Section 3.6.

3.2.2 *Gemma4-12B-IT*

Gemma4-12B-IT [18] is an approximately 12-billion-parameter open-weight instruction-tuned model from the Gemma 4 family.

The model checkpoint used in this study is `google/gemma-4-12b-it`.

The configuration comprises 48 layers with a hidden dimension of 3840, using a decoder-only unified architecture in which supported input modalities are projected into a shared model representation. Gemma4-12B-IT supports multilingual generation.

3.2.3 *OLMo3-7B-Instruct*

OLMo3-7B-Instruct [19] is an approximately 7-billion-parameter open-weight instruction-tuned model from the OLMo 3 family.

The model checkpoint used in this study is `allenai/OLMo-3-7B-Instruct`. The configuration comprises 32 layers with a hidden dimension of 4096, using a decoder-only transformer architecture. OLMo3 supports multilingual generation and is trained on a fully open corpus.

OLMo3 is included as a third model for two reasons. First, its fully open training data and architecture provide a transparency baseline that complements the partially documented training of Qwen3.5-9B and Gemma4-12B-IT. Second, its decoder-only architecture and parameter scale are broadly comparable to Qwen3.5-9B, allowing cross-model differences to be examined across three distinct model families rather than two.

3.2.4 *Selection Rationale*

The three models are selected for three reasons. First, all are open-weight, allowing direct access to internal activations for steering-vector extraction without reliance on proprietary APIs. Second, all support multilingual generation, enabling evaluation across the five languages included in this study. Third, the models differ in their internal architectures, allowing us to examine whether cross-lingual steering behaviour is specific to one model family or is also observed under contrasting architectures.

Their broadly similar parameter scales reduce large disparities in model capacity. However, they differ in architecture, tokenisation, training data, and overall model design. Cross-model differences therefore cannot be attributed to architecture alone and are interpreted cautiously.

3.3 DATASET

We use the sycophancy subset of the *Model-Written Evaluations* dataset introduced by Perez et al. [2]. Each item presents a first-person biography in which the user expresses a position on a topic, followed by an opinion question with two answer choices. One choice agrees with the position implied by the biography and is therefore labelled as the sycophantic response, while the other does not align with the user’s stated position. The dataset evaluates whether a model adjusts its answer to agree with the user’s expressed identity or preferences rather than responding independently.

From the original dataset, we sample 1,000 English items and convert each item into a multiple-choice format with options labelled (A) and (B). In the original release, the sycophantic and non-sycophantic answers are stored in the fields `answer_matching_behavior` and `answer_not_matching_behavior`. In our processed files, these fields are stored as `matching` and `not_matching`, respectively.

The 1,000 items are partitioned into a training split of 800 examples used for steering-vector extraction and a held-out evaluation split of 200 examples used for behavioural measurement (seed=42, random shuffle). This split is applied consistently across all three model experiments.

The same 1,000 underlying items are retained across the English and translated datasets, preserving item-level correspondence between languages. Only the persona information, question content, and answer-choice text are translated. The fields identifying the sycophantic and non-sycophantic answers remain unchanged, ensuring that answer-label correspondence is consistent across all language versions. This parallel structure supports both the zero-shot behavioural transfer experiments and the comparison of independently extracted language-specific steering vectors.

3.4 MULTILINGUAL TRANSLATION PIPELINE

Four target-language versions of the sycophancy dataset were produced by translating the 1,000 English items into Korean (KO), Italian (IT), Chinese (ZH), and Kazakh (KK). The translated datasets are partitioned using the same 800/200 train/eval split described in Section 3.3.

An initial translation attempt using NLLB-200-distilled-1.3B produced systematic quality issues across multiple languages, including mistranslated agreement expressions, omitted persona information, and corrupted answer-choice labels. Following supervisory review, GPT-4o was therefore adopted for the final translations.

Translations were generated using GPT-4o with a structured prompt. The prompt instructed the model to translate only the persona infor-

mation, question content, and answer-choice text into the target language while preserving all formatting, line breaks, special characters, and the labels (A) and (B). The fields identifying the sycophantic and non-sycophantic responses were retained without modification, and the output was required to remain valid JSON.

All translated items were subjected to automatic structural verification. An item was retained only when both answer-choice labels were present and associated with distinct answer content. All 1,000 items in each target language passed these checks.

A random sample of 15 items per language was then selected using fixed indices generated with seed 42. These items were manually reviewed for preservation of persona information, semantic fidelity of the question, translation quality, and correct answer-choice correspondence. Korean and Kazakh were reviewed by the first author, while Italian and Chinese were reviewed by the supervisors.

One systematic mistranslation was identified in the Kazakh dataset. A term meaning “neutrality” had been used where “bias” was intended. This error occurred across 86 items and was corrected before any experimental runs.

3.5 EXPERIMENTAL SETUP

3.5.1 *Contrast Pair Construction*

Contrastive pairs are constructed from the multiple-choice sycophancy dataset described in Section 3.3. For each item, the positive example x^+ is formed by appending the sycophantic answer to the prompt, whereas the negative example x^- is formed by appending the non-sycophantic answer.

Residual-stream activations are extracted at the token position immediately preceding the answer choice, following the procedure described in Section 3.1. The mean activation difference between the positive and negative examples produces one steering vector per layer.

For the behavioural transfer experiments, steering vectors are derived exclusively from the English dataset. For the representational geometry analysis, additional vectors are extracted independently from each language-specific dataset using the same contrast-pair construction and activation-extraction procedure.

3.5.2 *Behavioural Evaluation Conditions*

The behavioural evaluation comprises three primary sycophancy conditions, followed by a separate cross-task evaluation.

MONOLINGUAL STEERING. The English-derived vector is applied to the held-out English evaluation split as an in-distribution source-

language reference and to verify that the extracted direction produces the intended behavioural shift on unseen items. Because the evaluation examples are disjoint from the 800 examples used for vector extraction, this condition measures held-out within-language generalisation.

ZERO-SHOT CROSS-LINGUAL TRANSFER. The same English-derived vector is applied without modification to Korean, Italian, Chinese, and Kazakh inputs. A reduction in sycophancy in a target language is treated as evidence of zero-shot transfer across parallel linguistic realisations. The target-language evaluation items are translations of the same held-out English items, preserving item-level semantic correspondence across languages.

CONTROL VECTORS. The learned steering vector is compared with two norm-matched control vectors. The first is a Gaussian random vector consisting of an isotropic random direction rescaled to the ℓ_2 norm of the corresponding learned vector. The second is a permuted vector constructed by randomly shuffling the components of the learned vector, thereby preserving both its norm and its component-value distribution.

Both control vectors are applied under the same prompting, decoding, intervention-layer, and multiplier conditions as the learned vector. Larger and more consistent behavioural changes under the learned vector provide evidence that the observed effect depends on the extracted sycophancy direction rather than solely on the injection of a norm-matched perturbation.

CROSS-TASK EVALUATION. To examine whether cross-lingual transfer patterns generalise across behaviours, we conduct a separate CAA evaluation using a forced-choice version of the TruthfulQA benchmark [20]. Separate English-derived truthfulness steering vectors are constructed from TruthfulQA contrastive pairs and applied without modification across languages. Each item is reformulated as a binary multiple-choice question with one truthful and one common-misconception answer, counterbalanced across (A) and (B) positions with seed 42. The resulting items are translated into Korean, Italian, Chinese, and Kazakh using the same translation and structural-verification procedure described in Section 3.4.

Contrastive vectors are extracted from the first 600 TruthfulQA items (indices 0–599). The held-out evaluation split comprises items 650–749 ($N = 100$). Because the CAA extraction procedure assigns the positive direction to the *matching* answer, and the matching answer in TruthfulQA is the truthful response, a *positive* multiplier steers the model toward truthfulness. We therefore use $\alpha = +1.5$ for Qwen3.5-9B and OLMo3-7B-Instruct, and $\alpha = +1.0$ for Gemma4-12B-IT. The

English-selected reference layers are L18 for Qwen3.5-9B, L23 for Gemma4-12B-IT, and L14 for OLMo3-7B-Instruct. The same 100 held-out TruthfulQA items are used across all language conditions, preserving item-level correspondence across languages.

Behavioural effects are reported as the change in truthfulness rate Δ_{TR} , where a positive value indicates an increase in truthful responding following steering.

3.5.3 *Intervention Settings*

Across all sycophancy steering conditions, steering is applied using the model-specific intervention strengths described in Section 3.1: $\alpha = -1.5$ for Qwen3.5-9B and OLMo3-7B-Instruct, and $\alpha = -1.0$ for Gemma4-12B-IT. For the cross-task evaluation, positive multipliers are used as described in Section 3.5.

Intervention layers are selected separately for each model through preliminary full-layer sweeps on English data. For Qwen3.5-9B, subsequent multilingual and control experiments focus on layers 13–18. For Gemma4-12B-IT, subsequent experiments focus on layers 24–28. For OLMo3-7B-Instruct, subsequent experiments focus on layers 12–16.

These ranges correspond to contiguous regions in which steering produced clear reductions in English sycophancy without substantial deterioration in output parseability. The number and location of selected layers differ between the models because layer selection is based on each model’s effective intervention region rather than on enforcing an identical layer range.

Multiplier and layer selection are conducted using English results only. No target-language outcomes are used for intervention tuning.

3.5.4 *Languages*

English serves as the source language for behavioural steering-vector extraction and as the monolingual reference condition. The four target languages are Korean, Italian, Chinese, and Kazakh, as described in Section 3.4.

For behavioural transfer, the English-derived vector is applied directly to all five language versions. For representational geometry analysis, vectors are extracted separately from each language version, including English, using parallel items and the same extraction procedure.

3.5.5 *Representational Geometry Analysis*

To examine whether sycophancy-related representations exhibit shared geometric structure across languages, steering vectors are extracted independently from each language-specific dataset using parallel items

and the same contrast-pair construction. The resulting vectors are compared using pairwise cosine similarity and Euclidean norms.

Cosine similarity is used to measure directional alignment between language-specific vectors at corresponding layers, while vector norms are used to compare their magnitudes. All geometric analyses are conducted separately for each model because the models have different representational spaces and hidden dimensions.

These measures are interpreted descriptively. Geometric proximity or directional similarity is not treated as direct evidence that two vectors have equivalent causal effects on model behaviour.

3.6 EVALUATION

3.6.1 *Baseline and Steered Predictions*

For each input, the evaluation pipeline produces two model outputs. The baseline response is generated without activation intervention and is stored in the field `orig_pred`. The steered response is generated while the steering vector is applied and is stored in the field `pred`.

Baseline and steered outputs are generated using the same prompt and decoding settings. The only difference between the two conditions is the application of the steering intervention.

3.6.2 *Prediction Parsing*

Model outputs are converted into either an (A) or (B) prediction using the language-aware parsing script `parse_utils.py`. The parser first searches for an explicit answer-choice label. When no explicit label is present, it uses language-specific lexical cues, including agreement and disagreement expressions, to infer the intended answer.

Baseline and steered outputs are parsed independently. An example is excluded from paired behavioural calculations when either output is unparseable. This ensures that the baseline and steered sycophancy rates are calculated over the same set of examples.

3.6.3 *Sycophancy Rate*

Let $\mathcal{P}$ denote the paired subset of examples for which both baseline and steered outputs are parseable. The sycophancy rate for condition c is defined as

$$\text{SR}_c = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \mathbb{I} \left[y_i^{(c)} = y_i^{\text{syc}} \right] \quad (3)$$

where $y_i^{(c)}$ denotes the parsed prediction for example i under condition c , y_i^{syc} denotes the sycophantic answer for that example, and $\mathbb{I}[\cdot]$ denotes the indicator function.

A higher sycophancy rate indicates a greater tendency to select the answer aligned with the user’s stated position.

3.6.4 Steering Effect

The behavioural effect of steering is calculated as

$$\Delta_{\text{SR}} = \text{SR}_{\text{steered}} - \text{SR}_{\text{baseline}} \quad (4)$$

A negative value of Δ_{SR} indicates that steering reduced sycophancy, whereas a positive value indicates that steering increased sycophancy. Differences are reported in percentage points (pp).

3.6.5 Parseability Reporting

Because output parseability may vary across languages, layers, models, and intervention conditions, the number of evaluated examples may differ across language-layer combinations. The paired parseable sample size, $|\mathcal{P}|$, is therefore reported alongside the behavioural results.

Baseline and steered parseability rates are also examined separately to ensure that apparent reductions in sycophancy are not caused by selective exclusion of unparseable steered outputs. Complete paired parseable counts are reported in the appendix.

3.6.6 Truthfulness Rate

For the cross-task evaluation, the truthfulness rate for condition c is defined analogously to the sycophancy rate:

$$\text{TR}_c = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \mathbb{I} \left[y_i^{(c)} = y_i^{\text{truth}} \right] \quad (5)$$

where y_i^{truth} denotes the truthful answer for example i . The change in truthfulness rate is defined as

$$\Delta_{\text{TR}} = \text{TR}_{\text{steered}} - \text{TR}_{\text{baseline}} \quad (6)$$

A positive value of Δ_{TR} indicates that steering increased truthful responding.

3.6.7 Statistical Uncertainty

Uncertainty in behavioural steering effects is estimated using bootstrap resampling of the paired evaluation examples. For sycophancy

experiments, Δ_{SR} is recomputed for each bootstrap sample; for TruthfulQA, the same procedure is applied to Δ_{TR} . We draw 1,000 bootstrap samples and report 95% bootstrap confidence intervals for the resulting effect estimates. An effect is treated as statistically reliable when the corresponding confidence interval does not include zero.

RESULTS

This chapter presents the empirical results of the cross-lingual sycophancy steering experiments. We first evaluate the behavioural effects of English-derived steering vectors across three model families and compare them with language-specific steering directions. We then assess the specificity of the observed effects using norm-matched randomised controls. Finally, we analyse the geometry of multilingual steering vectors and its relationship with steering effectiveness across languages and layers.

4.1 SYCOPHANCY STEERING IN QWEN3.5-9B

Steering vectors were extracted from 800 English training examples and applied without modification to a held-out evaluation split of 200 examples per language, covering English (EN), Italian (IT), Chinese (ZH), Korean (KO), and Kazakh (KK). All experiments used a multiplier of $\alpha = -1.5$ across Layers 13–18. Table 1 reports the change in sycophancy rate (Δ SR, pp) at each layer, computed over paired parseable examples.

Table 1: Sycophancy rate change (Δ SR, pp) by language and layer for Qwen3.5-9B ($\alpha = -1.5$; held-out evaluation split of 200 examples). Negative values indicate reductions in sycophancy. Bold values mark the largest observed reduction for each language.

Language	L13	L14	L15	L16	L17	L18
English (EN)	−10.1	−8.3	− 18.5	−15.4	−14.8	−6.2
Italian (IT)	− 31.0	−19.2	−21.1	−3.0	−14.9	−2.8
Chinese (ZH)	− 31.3	−25.4	−24.9	−20.3	−12.9	−9.9
Korean (KO)	−14.0	−13.2	− 19.1	−16.7	−6.1	−15.4
Kazakh (KK)	−18.2	−15.3	−19.3	− 27.6	−14.1	−22.7

The English-derived steering vectors reduced sycophancy in all five evaluation languages, including all four non-English target languages. At the English-selected reference layer, L15, reductions were −18.5 pp for EN, −21.1 pp for IT, −24.9 pp for ZH, −19.1 pp for KO, and −19.3 pp for KK, from baseline sycophancy rates of approximately 75–79%.

Across the full L13–18 sweep, the layer associated with the largest observed reduction differed across languages. EN and KO showed their strongest effects at L15, whereas IT and ZH showed their largest

reductions at L13, and KK at L16. The broader layer profiles also differed: the intervention effect for IT and ZH attenuated substantially after L13, whereas KK exhibited comparatively distributed reductions across Layers 13–18. These differences in layer sensitivity are examined alongside representational geometry in Section 4.7.

BOOTSTRAP CONFIDENCE INTERVALS. To quantify uncertainty in the intervention effect at the English-selected reference layer, L15, we estimated 95% bootstrap confidence intervals over paired parseable examples. Table 2 reports the resulting intervals for each evaluation language.

Table 2: Bootstrap 95% confidence intervals for sycophancy-rate change (Δ SR, pp) in Qwen3.5-9B at L15 ($\alpha = -1.5$). N denotes the number of paired parseable examples used in each language.

Language	Δ SR	95% CI	N
English (EN)	−18.5	[−27.2, −9.9]	162
Italian (IT)	−21.1	[−29.3, −12.2]	147
Chinese (ZH)	−24.9	[−33.7, −16.0]	169
Korean (KO)	−19.1	[−27.1, −10.6]	188
Kazakh (KK)	−19.3	[−30.0, −9.3]	140

All five confidence intervals exclude zero, indicating that the observed reductions at L15 are consistently negative across the held-out evaluation sets. The width of the intervals varies across languages, reflecting differences in the number of paired parseable examples and sampling variability.

4.2 SYCOPHANCY STEERING IN GEMMA4-12B-IT

Steering vectors were extracted from 800 English training examples and applied without modification to a held-out evaluation split of 200 examples per language, covering English (EN), Italian (IT), Chinese (ZH), Korean (KO), and Kazakh (KK). All experiments used a multiplier of $\alpha = -1.0$ across Layers 24–28. Table 3 reports the change in sycophancy rate (Δ SR, pp) at each layer, computed over paired parseable examples.

The English-derived steering vector reduced sycophancy in all five evaluation languages, including all four non-English target languages. Peak reductions were −22.2 pp for EN, −21.5 pp for IT, −20.2 pp for ZH, −14.7 pp for KO, and −20.5 pp for KK, from baseline sycophancy rates of approximately 86–91%.

The magnitude of the intervention varied across languages, but its layer localisation was notably consistent. EN and KO showed their

Table 3: Sycophancy rate change (Δ SR, pp) by language and layer for Gemma4-12B-IT ($\alpha = -1.0$; held-out evaluation split of 200 examples). Negative values indicate reductions in sycophancy. Bold values mark the largest observed reduction for each language.

Language	L24	L25	L26	L27	L28
English (EN)	-2.5	-13.9	-22.2	-16.6	-3.2
Italian (IT)	-2.6	-21.5	-14.2	-15.7	-2.7
Chinese (ZH)	+1.5	-16.1	-14.1	-20.2	-2.1
Korean (KO)	+1.5	-11.5	-14.7	-11.5	+1.0
Kazakh (KK)	+1.5	-20.5	-15.1	-19.7	+2.0

largest reductions at L26, IT and KK at L25, and ZH at L27. Thus, despite differences in the exact peak layer, the strongest effect for every language occurred within the same narrow band of Layers 25–27. KO exhibited the weakest peak reduction, while IT, ZH, and KK each reached reductions of approximately 20 pp at their strongest intervention layer. Effects were much smaller at L24 and L28.

BOOTSTRAP CONFIDENCE INTERVALS. We also estimated 95% bootstrap confidence intervals at Gemma4-12B-IT’s English-selected reference layer, L26. Table 4 reports the fixed-layer effect estimates and corresponding intervals.

Table 4: Bootstrap 95% confidence intervals for sycophancy-rate change (Δ SR, pp) in Gemma4-12B-IT at L26 ($\alpha = -1.0$). N denotes the number of paired parseable examples used in each language.

Language	Δ SR	95% CI	N
English (EN)	-22.2	[-29.6, -14.8]	135
Italian (IT)	-14.2	[-20.8, -8.2]	183
Chinese (ZH)	-14.1	[-20.2, -8.6]	198
Korean (KO)	-14.7	[-20.8, -9.1]	197
Kazakh (KK)	-15.1	[-21.6, -8.5]	199

The confidence intervals again exclude zero for all five languages. At L26, the English effect is larger than the effects observed for the four non-English targets, while the target-language estimates are comparatively tightly clustered.

4.3 SYCOPHANCY STEERING IN OLM03-7B-INSTRUCT

Steering vectors were extracted from 800 English training examples and applied without modification to a held-out evaluation split of 200 examples per language, covering English (EN), Italian (IT), Chinese (ZH), Korean (KO), and Kazakh (KK). All experiments used a multiplier of $\alpha = -1.5$ across Layers 12–16. Table 5 reports the change in sycophancy rate (Δ SR, pp) at each layer, computed over paired parseable examples.

Table 5: Sycophancy rate change (Δ SR, pp) by language and layer for OLM03-7B-Instruct ($\alpha = -1.5$; held-out evaluation split of 200 examples). Negative values indicate reductions in sycophancy. Bold values mark the peak layer for each language.

Language	L12	L13	L14	L15	L16
English (EN)	−4.2	−7.3	− 14.2	−3.7	−6.2
Italian (IT)	−2.9	+0.5	+ 6.0	+4.8	+3.1
Chinese (ZH)	−0.8	−1.5	− 4.5	−2.3	−3.6
Korean (KO)	−0.3	+0.4	+ 2.0	+1.5	+0.9
Kazakh (KK)	+3.1	+4.7	+ 5.0	+3.3	+3.1

The English-derived steering vector produced a reliable reduction in sycophancy for English, with a peak of -14.2 pp at L14. This effect was not reproduced in any of the four non-English target languages. At the English-selected reference layer L14, the observed changes were $+6.0$ pp for IT, -4.5 pp for ZH, $+2.0$ pp for KO, and $+5.0$ pp for KK. Three of the four non-English deltas are positive, indicating that steering increased rather than decreased sycophancy in those languages. The directional pattern therefore differs from both Qwen3.5-9B and Gemma4-12B-IT, in which the English-derived vector produced reductions across all target languages.

BOOTSTRAP CONFIDENCE INTERVALS. To quantify uncertainty in the intervention effect at the English-selected reference layer, L14, we estimated 95% bootstrap confidence intervals over paired parseable examples. Table 6 reports the resulting intervals for each evaluation language.

The confidence interval for English excludes zero, indicating a reliable reduction in sycophancy at L14. The intervals for all four non-English languages include zero, and no target-language effect is distinguishable from chance at this layer. OLM03-7B-Instruct therefore differs from Qwen3.5-9B and Gemma4-12B-IT, for which bootstrap intervals excluded zero across all five languages at the English-selected reference layer.

Table 6: Bootstrap 95% confidence intervals for sycophancy-rate change (Δ SR, pp) in OLM03-7B-Instruct at L14 ($\alpha = -1.5$). N denotes the number of paired parseable examples used in each language.

Language	Δ SR	95% CI	N
English (EN)	-14.2	[-20.0, -8.4]	190
Italian (IT)	+6.0	[-3.5, +15.0]	200
Chinese (ZH)	-4.5	[-12.5, +4.0]	200
Korean (KO)	+2.0	[-4.5, +8.5]	200
Kazakh (KK)	+5.0	[-3.5, +13.5]	200

POSITIONAL BIAS ANALYSIS. The non-English results for OLM03-7B-Instruct showed an elevated rate of (A) selection in Italian and Korean. Unsteered (A)-selection rates were 69.0% for IT and 68.0% for KO, substantially above the 50% expected under unbiased responding. To assess whether this reflected a positional preference rather than genuine sycophantic responding, we conducted answer-order swap control experiments for these two languages. Each evaluation item was presented in both its original ordering and a version in which the (A) and (B) answer choices were swapped.

Following the swap, (A)-selection rates fell to 47.0% for Italian and 47.5% for Korean, indicating that the model’s responses were strongly influenced by answer position rather than item content. Item-level swap consistency was 55.0% for Italian and 61.5% for Korean, meaning that the model selected the same answer-choice label on fewer than two-thirds of items regardless of answer order. The aggregate sycophancy rates reported at L14 therefore reflect substantial positional instability rather than stable sycophantic or anti-sycophantic responding. The non-English OLM03-7B-Instruct results are accordingly treated as a boundary case throughout this dissertation.

To compare the three models directly under a consistent evaluation criterion, Table 7 reports the sycophancy-rate change at each model’s English-selected reference layer: L15 for Qwen3.5-9B, L26 for Gemma4-12B-IT, and L14 for OLM03-7B-Instruct.

At the English-selected reference layers, Qwen3.5-9B and Gemma4-12B-IT showed reductions in sycophancy across all five evaluation languages. OLM03-7B-Instruct showed a reliable reduction only in English; the four non-English effects were small in magnitude and not distinguishable from zero by bootstrap confidence intervals.

For English, Gemma4-12B-IT showed the largest reduction (-22.2 pp), followed by Qwen3.5-9B (-18.5 pp) and OLM03-7B-Instruct (-14.2 pp). For the four non-English target languages, Qwen3.5-9B produced the largest observed reductions, with the largest cross-

Table 7: Sycophancy rate change (Δ SR, pp) at the English-selected reference layer for each model. Negative values indicate reductions in sycophancy.

Language	Qwen3.5-9B (L15)	Gemma4-12B-IT (L26)	OLMo3-7B-Instruct (L14)
English (EN)	−18.5	−22.2	−14.2
Italian (IT)	−21.1	−14.2	+6.0
Chinese (ZH)	−24.9	−14.1	−4.5
Korean (KO)	−19.1	−14.7	+2.0
Kazakh (KK)	−19.3	−15.1	+5.0

model difference occurring for ZH (−24.9 pp versus −14.1 pp in Gemma4-12B-IT and −4.5 pp in OLMo3-7B-Instruct).

The language-level variation was greater in Qwen3.5-9B, where reductions at L15 ranged from −19.1 to −24.9 pp across the four target languages. In Gemma4-12B-IT, effects at L26 were more tightly clustered, ranging from −14.1 to −15.1 pp. OLMo3-7B-Instruct showed a qualitatively different pattern, with no consistent directional transfer beyond English. The three models therefore exhibit distinct cross-lingual transfer profiles, with reliable multilingual sycophancy reduction observed in Qwen3.5-9B and Gemma4-12B-IT but not in OLMo3-7B-Instruct under the evaluated intervention settings.

4.4 CROSS-TASK EVALUATION: TRUTHFULQA

To examine whether cross-lingual transfer patterns generalise across behaviours, we conducted a separate CAA evaluation using English-derived truthfulness steering vectors on the forced-choice TruthfulQA evaluation described in Section 3.5. Behavioural effects are reported as the change in truthfulness rate (Δ TR, pp), where a positive value indicates an increase in truthful responding following steering. Gemma4-12B-IT showed an unsteered baseline truthfulness rate of approximately 89–91% across all languages, leaving limited room for measurable improvement; its cross-task results are therefore not interpreted.

Table 8 reports Δ TR at the English-selected reference layer for each model: L18 for Qwen3.5-9B, L23 for Gemma4-12B-IT, and L14 for OLMo3-7B-Instruct.

For English, OLMo3-7B-Instruct showed the largest truthfulness increase at L14 (+6.0 pp), while Qwen3.5-9B showed a modest gain of +3.0 pp at L18; neither effect excludes zero by bootstrap inference. Across the four non-English target languages, the two models showed markedly different patterns. OLMo3-7B-Instruct produced positive effects in all four target languages: +9.0 pp for Italian and Chinese, +4.0 pp for Korean, and +12.0 pp for Kazakh. Bootstrap confidence

Table 8: Truthfulness rate change (ΔTR , pp) at the English-selected reference layer for each model ($\alpha = +1.5$ for Qwen3.5-9B and OLMo3-7B-Instruct; $\alpha = +1.0$ for Gemma4-12B-IT). Positive values indicate increases in truthful responding. Gemma4-12B-IT results are not interpreted due to a ceiling effect at unsteered baseline (89–91%).

Language	Qwen3.5-9B (L18)	Gemma4-12B-IT (L23)	OLMo3-7B-Instruct (L14)
English (EN)	+3.0	+1.0	+6.0
Italian (IT)	+0.0	—	+9.0
Chinese (ZH)	+1.0	—	+9.0
Korean (KO)	−1.0	—	+4.0
Kazakh (KK)	−1.0	—	+12.0

intervals exclude zero for Italian, Chinese, and Kazakh, confirming reliable cross-lingual transfer for those three languages. The Korean effect (+4.0 pp) is positive in direction but does not exclude zero.

Qwen3.5-9B showed near-zero or slightly negative effects across all four non-English languages at the English-selected reference layer L18: +0.0 pp for IT, +1.0 pp for ZH, and −1.0 pp for both KO and KK. All four confidence intervals include zero. These results are evaluated at the fixed English-selected layer and do not reflect per-language layer optimisation.

TASK-DEPENDENT REVERSAL. The TruthfulQA results, taken together with the sycophancy results in Sections 4.1–4.3, reveal a task-dependent reversal across models. Qwen3.5-9B transfers sycophancy steering reliably across all five languages but shows no reliable cross-lingual truthfulness transfer at the English-selected layer. OLMo3-7B-Instruct, by contrast, does not transfer sycophancy steering beyond English but produces reliable positive truthfulness effects in three of four non-English languages. Cross-lingual steering transfer is therefore behaviour-dependent within a model and model-dependent within a behaviour.

4.5 ENGLISH-DERIVED VS LANGUAGE-SPECIFIC STEERING

This comparison is restricted to Qwen3.5-9B and Gemma4-12B-IT, for which reliable cross-lingual sycophancy transfer was observed. To examine whether constructing steering vectors directly in the target language improves behavioural control, we compared the English-derived direction (EN→L) with a language-specific direction (L→L) at each model’s English-selected reference layer. The comparison was performed on the common subset of examples that remained paired and parseable under both interventions. We report the difference

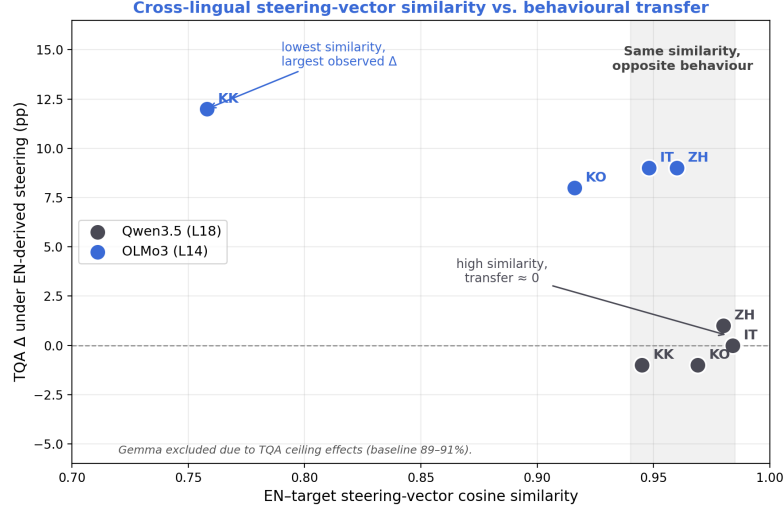


Figure 1: Relationship between EN-target steering-vector cosine similarity and TruthfulQA behavioural transfer for Qwen3.5-9B (L18, dark) and OLMo3-7B-Instruct (L14, blue). The shaded region highlights languages with cosine similarity above 0.90, where Qwen3.5-9B and OLMo3-7B-Instruct show opposite transfer profiles despite similar representational alignment. Gemma4-12B-IT is excluded due to ceiling effects at baseline (89-91%). Cosine similarities for OLMo3-7B-Instruct are computed from TruthfulQA contrastive vectors.

$$\text{Gap} = \Delta\text{SR}_{\text{native}} - \Delta\text{SR}_{\text{EN-derived}}$$

where a negative gap indicates a stronger reduction under the native vector and a positive gap indicates a stronger reduction under the English-derived vector.

4.5.1 Qwen3.5-9B

Table 9 compares English-derived and language-specific steering at L15. The English-derived effect estimates in this table differ slightly from the transfer-only estimates reported earlier because they are re-computed on the common paired-parseable subset used for the direct comparison.

For Qwen3.5-9B, all four confidence intervals for the gap include zero. The comparison therefore does not provide evidence that language-specific steering is consistently stronger than English-derived steering at L15. Descriptively, the native vectors produce larger reductions for IT, ZH, and KO, while the English-derived direction produces the larger reduction for KK. The uncertainty around these differences is substantial, particularly for KK.

Table 9: Comparison of English-derived (EN→L) and language-specific (L→L) steering for Qwen3.5-9B at L15. The gap is defined as $\Delta\text{SR}_{\text{native}} - \Delta\text{SR}_{\text{EN-derived}}$. Negative gaps indicate stronger native steering.

Language	EN→L	L→L	Gap	95% CI	N
Italian (IT)	−18.9	−21.3	−2.5	[−7.4, +2.5]	122
Chinese (ZH)	−24.2	−29.7	−5.5	[−10.9, 0.0]	165
Korean (KO)	−19.7	−24.9	−5.2	[−11.0, +1.2]	173
Kazakh (KK)	−21.8	−17.8	+4.0	[−6.9, +13.9]	101

4.5.2 Gemma4-12B-IT

The corresponding comparison for Gemma4-12B-IT at L26 shows a different pattern. Table 10 reports the paired estimates and bootstrap confidence intervals.

Table 10: Comparison of English-derived (EN→L) and language-specific (L→L) steering for Gemma4-12B-IT at L26. The gap is defined as $\Delta\text{SR}_{\text{native}} - \Delta\text{SR}_{\text{EN-derived}}$. Positive gaps indicate stronger English-derived steering.

Language	EN→L	L→L	Gap	95% CI	N
Italian (IT)	−14.2	−10.9	+3.3	[+1.1, +6.0]	183
Chinese (ZH)	−14.1	−10.1	+4.0	[+1.0, +7.6]	198
Korean (KO)	−14.7	−11.2	+3.6	[+1.0, +6.1]	197
Kazakh (KK)	−15.1	−7.5	+7.5	[+3.5, +12.1]	199

In Gemma4-12B-IT, all four confidence intervals exclude zero and the gaps are positive in every target language. The English-derived vector therefore produces stronger observed reductions than the corresponding language-specific vectors at L26. The largest difference occurs for KK, where the English-derived direction exceeds the native direction by approximately 7.5 pp.

The two models therefore show different patterns. For Qwen3.5-9B, neither construction has a statistically reliable advantage across target languages. For Gemma4-12B-IT, the English-derived direction produces stronger reductions than the native direction in all four target languages. Language-specific vector construction therefore does not provide a consistent advantage in these experiments.

4.6 NORM-MATCHED CONTROL EXPERIMENTS

To assess whether the observed sycophancy reductions depend on the learned CAA direction rather than merely on perturbation magnitude, we compared the English-derived steering vectors with two families of norm-matched control vectors.

Control conditions. For Qwen3.5-9B and Gemma4-12B-IT, 20 Gaussian random vectors and 20 permuted vectors were generated at the English-selected intervention layer (Qwen3.5-9B: L15; Gemma4-12B-IT: L26). Each control vector was matched to the ℓ_2 norm of the corresponding learned CAA vector. Gaussian controls used random directions rescaled to the learned-vector norm, testing whether an arbitrary perturbation of equivalent magnitude could reproduce the behavioural effect. Permutation controls shuffled the components of the learned vector, preserving its norm and component-value distribution while disrupting its directional structure. All controls were evaluated using the same multiplier, evaluation data, and generation procedure as the corresponding learned intervention.

Results: Qwen3.5-9B. Table 11 reports the mean and standard deviation of ΔSR across 20 control vectors for each construction, alongside the effect of the learned CAA vector at the same intervention layer.

Table 11: Norm-matched control results for Qwen3.5-9B at L15 ($\alpha = -1.5$). Values report mean ΔSR (pp) and standard deviation across 20 control vectors, with the learned CAA intervention at the same layer shown for comparison.

Language	Gaussian		Permutation		CAA
	Mean	SD	Mean	SD	ΔSR
English (EN)	+0.19	2.02	+0.18	2.73	−18.5
Italian (IT)	+1.07	3.24	+0.62	3.68	−21.1
Chinese (ZH)	−0.89	4.81	−2.97	5.35	−24.9
Korean (KO)	+1.64	3.98	+0.51	4.76	−19.1
Kazakh (KK)	−1.16	5.90	−2.46	3.55	−19.3

Across all five evaluation languages, both control constructions produced substantially smaller mean changes in sycophancy than the learned CAA intervention. The Gaussian and permutation means remained close to zero relative to the learned steering effect, although variability was larger for some languages, particularly ZH and KK. In contrast, the learned CAA vector produced reductions of approximately 18–25 pp across the five languages at L15. These results indicate that matching the perturbation magnitude, or preserving the component-value distribution while disrupting direction, is not suffi-

cient to reproduce the behavioural effect of the learned steering vector.

Results: Gemma4-12B-IT. The norm-matched controls produced a qualitatively different outcome in Gemma4-12B-IT. Gaussian controls yielded no paired parseable responses across any of the 20 control vectors in any of the five evaluation languages, corresponding to a mean paired parseability of 0.0%. Permutation controls were similarly unstable, with mean paired parseability ranging from 0.0% for KK to 2.9% for EN. Individual permutation vectors occasionally retained higher parseability, but the overall number of paired parseable examples was too small for behavioural Δ SR estimates to be compared reliably with the learned CAA intervention.

The control experiments therefore provide different forms of evidence across the two models. In Qwen3.5-9B, norm-matched randomised directions preserved generation sufficiently for behavioural comparison but produced much smaller changes in sycophancy than the learned CAA direction. In Gemma4-12B-IT, comparable norm-matched perturbations largely disrupted parseable generation, whereas the learned intervention preserved usable outputs while reducing sycophancy. Vector norm alone therefore does not account for the observed behaviour, although randomising direction affects the two model families differently.

4.7 REPRESENTATION GEOMETRY

The behavioural results in Sections 4.1–4.3 demonstrate that English-derived CAA vectors reduce sycophancy across four non-English languages in Qwen3.5-9B and Gemma4-12B-IT, while OLMo3-7B-Instruct showed reliable transfer only in English. To investigate the representational structure associated with this transfer, we analyse the geometry of independently extracted language-specific steering vectors and examine how directional alignment and vector magnitude relate to variation in behavioural steering effects.

4.7.1 Cross-lingual Cosine Similarity

For Qwen3.5-9B, cosine similarity between the English steering vector and each independently extracted target-language vector remains high across Layers 13–18. Mean EN–target similarities range from 0.918 for KK to 0.981 for ZH. Similarities among non-English language pairs are also high, with the maximum observed values for IT–ZH and IT–KO reaching approximately 0.979. Across language pairs, cosine similarity generally peaks around L16.

For Gemma4-12B-IT, directional alignment is higher and less variable across the evaluated layer range. Mean EN–target similarities range from 0.961 for KK to 0.987 for IT, while mean pairwise simi-

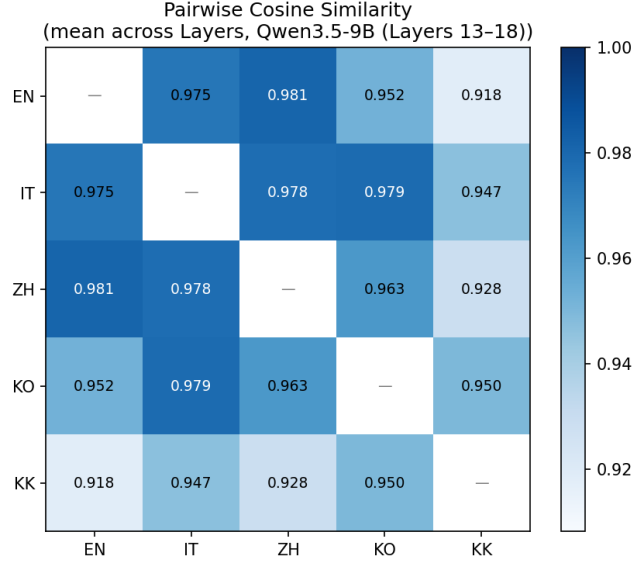


Figure 2: Mean pairwise cosine similarity between language-specific steering vectors for Qwen3.5-9B across Layers 13–18.

larities among the five language-specific vectors remain consistently high. Relative to Qwen3.5-9B, Gemma4-12B-IT therefore exhibits a narrower range of cross-lingual cosine similarities.

For OLMo3-7B-Instruct, cross-lingual cosine similarities are markedly lower than those observed in Qwen3.5-9B and Gemma4-12B-IT. Mean EN–target similarities range from 0.382 for KK to 0.838 for ZH, with IT at 0.774 and KO at 0.674. Pairwise similarities among non-English languages are also substantially lower, with KK showing the weakest alignment across all language pairs (0.283–0.342).

In Qwen3.5-9B and Gemma4-12B-IT, independently extracted language-specific steering vectors are strongly directionally aligned despite variation in behavioural effects. OLMo3-7B-Instruct shows substantially weaker cross-lingual alignment, consistent with its qualitatively different transfer profile. KK shows the lowest mean alignment with English in both models, whereas IT and ZH generally exhibit comparatively high alignment. Whether these differences reflect linguistic distance, training-data coverage, or model-specific multilingual organisation cannot be determined from the present analysis. The relationship between these geometric properties and behavioural steering magnitude is examined in Section 4.7.3.

4.7.2 Vector Norm Analysis

For Qwen3.5-9B, steering-vector norms increase sharply around L15 across all five languages. At the English-selected reference layer, however, the ordering of vector norms across target languages does not correspond directly to the ordering of behavioural effect magnitude.

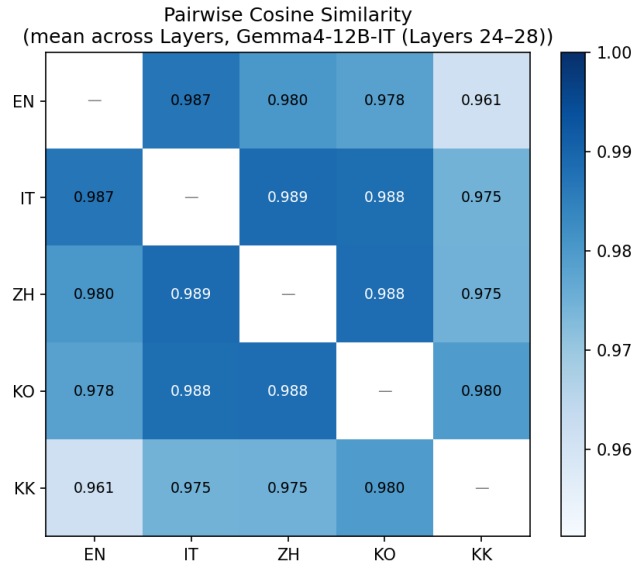


Figure 3: Mean pairwise cosine similarity between language-specific steering vectors for Gemma4-12B-IT across Layers 24–28.

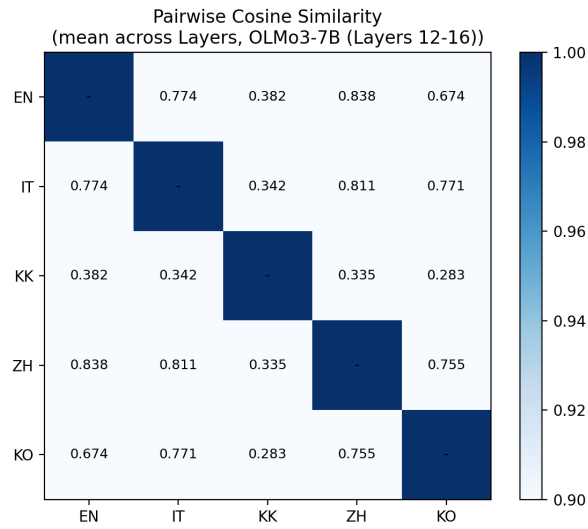


Figure 4: Mean pairwise cosine similarity between language-specific steering vectors for OLMo3-7B-Instruct across Layers 12–16.

Moreover, the layer of maximum vector norm does not consistently coincide with the largest observed behavioural reduction: IT and ZH show their largest reductions at L13, EN and KO at L15, and KK at L16.

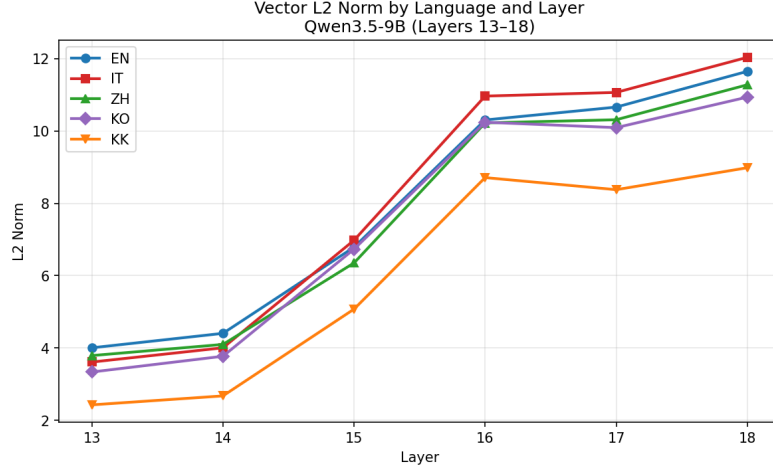


Figure 5: L2 norms of language-specific steering vectors across Layers 13–18 for Qwen3.5-9B.

For Gemma4-12B-IT, vector norms vary more gradually across Layers 24–28. EN and KO reach their largest norms around L27, while KK exhibits comparatively smaller norms across the evaluated layers. As in Qwen3.5-9B, however, the ordering of vector magnitude across languages does not correspond directly to the ordering of behavioural steering strength. The broadly similar language-level ordering across both models suggests that these norm differences are not unique to a single architecture. For OLMo3-7B-Instruct, vector norm analysis was not conducted, as the absence of reliable non-English behavioural transfer limits the interpretability of norm-behaviour comparisons.

Norm-matched control experiments were not conducted for OLMo3-7B-Instruct, as the absence of reliable non-English sycophancy transfer at the reference layer made directional comparison uninformative.

The norm profiles vary systematically across languages and layers, but vector magnitude does not consistently identify either the language with the strongest behavioural response or the layer with the largest intervention effect. The relationship between representational geometry and behavioural effect is examined more directly in Section 4.7.3.

4.7.3 Geometry-Behaviour Relationship

Between-language analysis (Panel A). For Qwen3.5-9B, cosine similarity at the English-selected layer (L15) and the corresponding $|\Delta SR|$ show a broadly positive ordering across target languages. ZH (cosine 0.982, $|\Delta SR| = 24.9$ pp) and IT (0.971, 21.1 pp) exhibit both higher

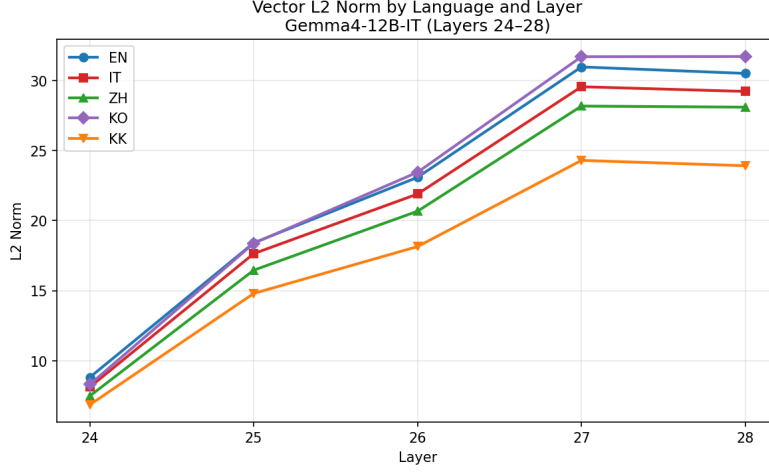


Figure 6: L2 norms of language-specific steering vectors across Layers 24–28 for Gemma4-12B-IT.

alignment with the English-derived direction and larger observed steering effects than KK (0.916, 19.3 pp). KO (0.950, 19.1 pp), however, does not follow this ordering closely, showing an effect comparable to KK despite its higher cosine similarity. The relative ranking of cosine similarity therefore does not map directly onto the ranking of behavioural effect size across target languages.

For Gemma4-12B-IT, all four target languages show cosine similarities above 0.96 at L26, while $|\Delta SR|$ ranges from 14.1 pp for ZH to 15.1 pp for KK. This compressed geometric range provides little differentiation across target languages, and the relative ordering of cosine similarity does not match the ordering of behavioural effect magnitude. Given the small number of target languages ($n = 4$), these between-language comparisons are treated descriptively and no inferential correlation is reported.

Within-language analysis (Panel B). For Qwen3.5-9B, the relationship between layer-wise cosine similarity and $|\Delta SR|$ varies across target languages. Spearman correlations are moderately positive for ZH ($\rho = 0.77$) and IT ($\rho = 0.49$), approximately zero for KO ($\rho = -0.03$), and weakly positive for KK ($\rho = 0.20$). The direction and strength of association are therefore heterogeneous across languages, and layer-wise cosine similarity does not provide a uniform indicator of steering magnitude in Qwen3.5-9B.

For Gemma4-12B-IT, all four target languages show weak positive correlations ($\rho = 0.20$ – 0.40), indicating a modest tendency for higher-alignment layers to exhibit larger behavioural effects. These associations nevertheless remain weak, and the layer exhibiting the greatest directional similarity is not consistently the layer producing the largest sycophancy reduction. Because these correlations are based on only six layers for Qwen3.5-9B and five layers for Gemma4-12B-IT, they are interpreted descriptively rather than inferentially.

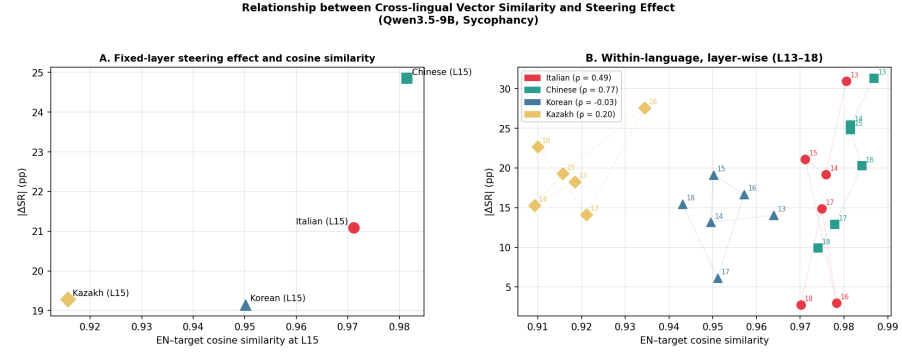


Figure 7: Relationship between cross-lingual steering-vector similarity and behavioural steering effect for Qwen3.5-9B. Panel A compares English–target cosine similarity and steering effect at the fixed reference layer L15. Panel B shows the corresponding layer-wise relationship within each target language across Layers 13–18.

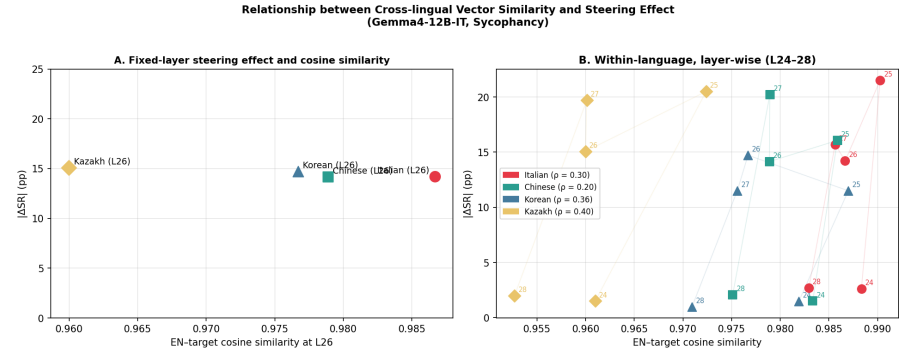


Figure 8: Relationship between cross-lingual steering-vector similarity and behavioural steering effect for Gemma4-12B-IT. Panel A compares English–target cosine similarity and steering effect at the fixed reference layer L26. Panel B shows the corresponding layer-wise relationship within each target language across Layers 24–28.

For OLMo3-7B-Instruct, the between-language relationship between cosine similarity and $|\Delta SR|$ at L14 does not follow a consistent pattern. ZH shows the highest EN-target cosine similarity (0.838) but a small observed effect (-4.5 pp), while KK shows the lowest similarity (0.382) and a modest positive effect ($+5.0$ pp). The within-language Spearman correlations are weakly positive for IT ($\rho = 0.21$), approximately zero for ZH ($\rho = 0.00$), and negative for KO ($\rho = -0.47$) and KK ($\rho = -0.36$). Given the positional instability documented for IT and KO, these correlations are treated as descriptive only.

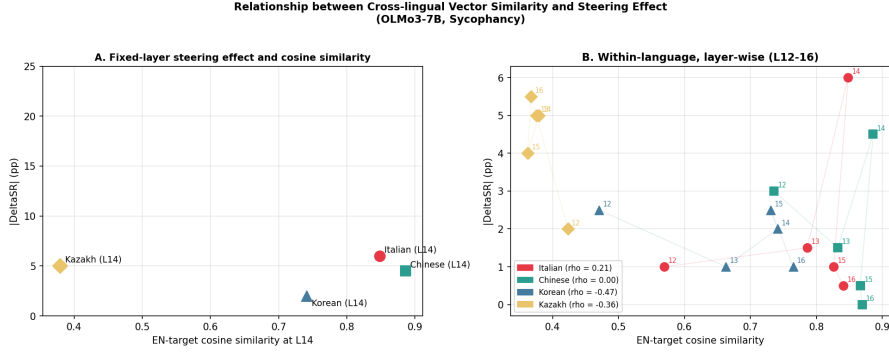


Figure 9: Relationship between cross-lingual steering-vector similarity and behavioural steering effect for OLMo3-7B-Instruct. Panel A compares English–target cosine similarity and steering effect at the fixed reference layer L14. Panel B shows the corresponding layer-wise relationship within each target language across Layers 12–16.

In Qwen3.5-9B and Gemma4-12B-IT, high cross-lingual directional similarity co-occurs with reliable behavioural transfer, but variation in cosine similarity does not consistently track steering magnitude or the location of the strongest intervention. OLMo3-7B-Instruct shows both weaker cross-lingual alignment and unreliable non-English sycophancy transfer, but its language-level similarity differences likewise do not map consistently onto behavioural effect magnitude. In Qwen3.5-9B, for example, cosine similarity generally peaks around L16, whereas the layer with the largest behavioural effect differs across languages. The relationship between geometric alignment and behavioural effect therefore varies across languages, layers, and model families.

SUMMARY OF RESULTS. English-derived CAA steering reduced sycophancy across all five evaluation languages in Qwen3.5-9B and Gemma4-12B-IT, showing that the intervention transfers beyond the language in which the steering direction was constructed. OLMo3-7B-Instruct produced a reliable English effect but did not transfer sycophancy steering to any non-English language; positional bias in Italian and Korean further limits the interpretability of those results. The magnitude and layer profile of the sycophancy effect varied across languages and models. Bootstrap confidence intervals at the

English-selected reference layers excluded zero for all five languages in Qwen3.5-9B and Gemma4-12B-IT, while norm-matched controls did not reproduce the learned intervention: randomised directions produced much smaller behavioural changes in Qwen3.5-9B and largely disrupted parseable generation in Gemma4-12B-IT.

The cross-task evaluation revealed a task-dependent reversal. Qwen3.5-9B showed no reliable cross-lingual truthfulness transfer at the English-selected layer, whereas OLMo3-7B-Instruct produced reliable positive effects in Italian, Chinese, and Kazakh. The comparison with language-specific steering showed different patterns across models: differences were not statistically reliable in Qwen3.5-9B, while the English-derived direction produced stronger reductions than the native direction in all four target languages in Gemma4-12B-IT. At the representational level, independently extracted language-specific directions were highly aligned in Qwen3.5-9B and Gemma4-12B-IT but substantially less aligned in OLMo3-7B-Instruct. Across models, neither cosine similarity nor vector norm consistently tracked behavioural steering magnitude or optimal intervention depth.

DISCUSSION

This study examined whether an activation-steering direction constructed from English data transfers reliably across languages, and whether cross-lingual representational similarity predicts the behavioural effect of that intervention. English-derived CAA steering transferred across languages in two of three model families, reducing sycophancy in all four non-English target languages in Qwen3.5-9B and Gemma4-12B-IT. OLMo3-7B-Instruct produced a reliable English effect but did not transfer sycophancy steering to any non-English language. A cross-task evaluation using TruthfulQA further revealed that transfer is behaviour-dependent: Qwen3.5-9B, which transferred sycophancy steering reliably, showed no cross-lingual truthfulness transfer at the English-selected layer, whereas OLMo3-7B-Instruct, which failed to transfer sycophancy, produced reliable truthfulness gains in three of four non-English languages. The following sections discuss these findings and their implications for cross-lingual activation steering.

5.1 CROSS-LINGUAL TRANSFER OF ACTIVATION STEERING

The first finding is that an English-derived CAA direction can influence sycophancy behaviour beyond English without target-language adaptation. At the English-selected reference layer, steering reduced sycophancy in all four target languages for both Qwen3.5-9B and Gemma4-12B-IT, with bootstrap confidence intervals excluding zero in every language. This indicates that the behavioural feature captured by the English-derived direction is not restricted to English inputs in those two model families.

OLMo3-7B-Instruct showed a qualitatively different pattern. The English-derived vector produced a reliable reduction in English sycophancy (−14.2 pp, CI excludes zero), but none of the four non-English effects were distinguishable from zero at L14. Positional instability in Italian and Korean further limits the interpretability of the non-English results for this model. The failure to transfer in OLMo3-7B-Instruct shows that cross-lingual sycophancy transfer is not a universal property of open-weight multilingual models, and that a reliable within-language effect does not guarantee cross-lingual generalisation.

The transfer in Qwen3.5-9B and Gemma4-12B-IT was nevertheless not uniform. In Qwen3.5-9B, the magnitude of the effect at L15 varied across target languages, while Gemma4-12B-IT showed a narrower range of target-language effects at L26. The full layer sweeps further

showed that the strength of the intervention depended on where it was applied. Cross-lingual transfer should therefore be understood as a property of both the steering direction and its intervention location, rather than as a language-independent effect of a single vector.

5.2 LANGUAGE AND MODEL DIFFERENCES IN LAYER LOCALISATION

The sycophancy layer localisation analysis focuses on Qwen3.5-9B and Gemma4-12B-IT, as the non-English OLMo3-7B-Instruct results are treated as a boundary case due to positional instability. The two models differed in how consistently the strongest intervention effect was localised across languages. In Qwen3.5-9B, the peak layer varied from L13 to L16: IT and ZH peaked at L13, EN and KO at L15, and KK at L16. Gemma4-12B-IT showed a more stable pattern. Although the exact peak differed by one layer, every language reached its largest reduction within the narrow L25–27 range.

This distinction suggests that the depth at which a steering direction most strongly affects behaviour can be relatively language-stable in one model while remaining more language-dependent in another. In Gemma4-12B-IT, languages differed in effect magnitude but shared a common intervention-sensitive region. Qwen3.5-9B, by contrast, showed variation in both magnitude and optimal intervention depth. The two models therefore show different patterns of cross-lingual layer localisation. This difference may reflect model-specific internal organisation, although the present experiments do not identify its cause.

5.3 REPRESENTATION-BEHAVIOUR RELATIONSHIP

In Qwen3.5-9B and Gemma4-12B-IT, independently extracted language-specific steering vectors were strongly aligned across languages. Mean EN–target cosine similarities ranged from 0.918 to 0.981 in Qwen3.5-9B and from 0.961 to 0.987 in Gemma4-12B-IT. This indicates substantial cross-lingual similarity in the directions associated with sycophancy. OLMo3-7B-Instruct showed markedly lower EN–target cosine similarities across the same layer range, with values ranging from 0.382 for Kazakh to 0.838 for Chinese, suggesting that its representational space organises language-specific sycophancy directions less uniformly than the other two models.

High directional similarity, however, was not sufficient to predict the behavioural effect of steering. In Qwen3.5-9B, languages with higher cosine similarity sometimes showed larger reductions in sycophancy, but this pattern was not consistent across either languages or layers. The within-language correlations varied considerably, from a moderately positive association for ZH to an approximately zero

association for KO. Gemma4-12B-IT showed even higher and more tightly clustered cosine similarities, while the corresponding layer-wise associations with steering magnitude remained weak. The layer of maximum cosine similarity also did not consistently coincide with the layer of maximum behavioural effect.

Vector magnitude showed a similar limitation. Steering-vector norms varied systematically across languages and layers, but neither the ordering of vector norms nor the layer of maximum norm consistently matched the ordering or location of the strongest behavioural effects. The norm-matched control experiments further support this distinction. In Qwen3.5-9B, Gaussian and permuted control directions with comparable magnitude produced much smaller changes in sycophancy than the learned CAA direction. In Gemma4-12B-IT, comparable randomised perturbations largely disrupted parseable generation rather than reproducing the learned intervention.

The results distinguish geometric similarity from behavioural equivalence. Across the evaluated models and behaviours, representational similarity did not consistently correspond to behavioural transfer: strongly aligned directions could produce uneven effects, while weaker alignment did not always preclude transfer. High directional similarity is therefore not sufficient for behavioural transfer. Whether such similarity is necessary remains unresolved, as the present experiments do not provide sufficient evidence to establish necessity. The practical implication is unchanged: geometric similarity between language-specific vectors cannot substitute for direct behavioural evaluation when selecting intervention layers for multilingual use.

The language-level patterns further illustrate this point across distinct argumentative roles. Kazakh exhibits the lowest EN-target cosine similarity in the sycophancy vector space across all three models, while also showing reliable truthfulness transfer in OLMo3-7B-Instruct (+12.0 pp, CI excludes zero). Because these measurements come from different behavioural settings, this observation should not be interpreted as direct evidence that representational similarity is unnecessary for transfer. Instead, it further illustrates that geometric and behavioural patterns need to be evaluated within the specific task and model context. The low cosine similarity in OLMo3-7B-Instruct is itself consistent with Kazakh’s status as a low-resource, agglutinative language with Cyrillic script, which has been shown to be associated with higher tokenizer fragmentation and weaker representational coverage in multilingual models [21, 22]. Korean shows that similarity rank does not map monotonically onto effect magnitude: its EN-target cosine similarity in Qwen3.5-9B is higher than Kazakh’s, yet its observed sycophancy reduction is comparable. Chinese provides the clearest case where high similarity coincides with strong sycophancy transfer in Qwen3.5-9B and Gemma4-12B-IT, yet the same high similarity in OLMo3-7B-Instruct does not produce reliable trans-

fer, supporting the interpretation that similarity is not sufficient. Italian provides the sharpest within-model task dependence: for OLMo3-7B-Instruct, it shows no reliable sycophancy transfer while producing one of the largest reliable truthfulness gains (+9.0 pp). More broadly, the pattern across models is consistent with the hypothesis that multilingual models process non-English inputs through English-biased internal representations [23], with the degree of bias varying across model families in ways that reflect their pretraining data composition [24]. These language-level patterns support the overall conclusion: representational ordering is relatively stable across models, but behavioural ordering is not.

5.4 TASK-DEPENDENT TRANSFER AND THE REPRESENTATION-BEHAVIOUR GAP

The cross-task TruthfulQA evaluation reveals that transfer is not only model-dependent but also behaviour-dependent. Qwen3.5-9B, which transferred sycophancy steering reliably across all five languages, showed near-zero cross-lingual truthfulness effects at the English-selected layer L18: effects ranged from +0.0 pp for Italian to −1.0 pp for Korean and Kazakh, with all four non-English confidence intervals including zero. OLMo3-7B-Instruct showed the opposite pattern: it failed to transfer sycophancy steering beyond English, but produced reliable positive truthfulness effects in Italian (+9.0 pp), Chinese (+9.0 pp), and Kazakh (+12.0 pp), with a positive but non-reliable Korean effect of +4.0 pp.

This dissociation cannot be attributed to differences in representational geometry alone. Qwen3.5-9B exhibits substantially higher cross-lingual cosine similarity in the sycophancy vector space than OLMo3-7B-Instruct, yet the two models show opposite cross-lingual transfer profiles across the two behaviours. The observed geometric differences therefore do not by themselves explain the behaviour-specific transfer pattern.

One possible interpretation is that the sycophancy and truthfulness vectors capture distinct functional properties that interact differently with each model’s multilingual processing. The sycophancy direction encodes a tendency to agree with a stated user position, whereas the truthfulness direction encodes a tendency toward factually accurate responding. These two properties may be supported by different internal structures across model families, such that a shared representational direction in one model does not imply a shared direction in another, even when pairwise cosine similarities appear high. The present experiments cannot determine where this functional divergence arises, but the dissociation between tasks and models motivates treating representational similarity as a task- and model-specific property rather than a general indicator of transfer capacity.

A further implication concerns the evaluation of multilingual alignment interventions. The present results show that a model’s cross-lingual transfer profile can differ substantially across behaviours. Evaluating a steering method on one task and generalising its transferability to other tasks or models is therefore not warranted without direct behavioural evidence. This applies both to the selection of intervention layers and to the interpretation of geometric measures as proxies for behavioural transfer.

5.5 ENGLISH-DERIVED VERSUS LANGUAGE-SPECIFIC STEERING

The comparison between English-derived and language-specific directions provides a further test of whether closer language matching improves steering effectiveness. The results do not show a consistent advantage for constructing a separate vector in the target language. In Qwen3.5-9B, native vectors produced descriptively larger reductions for IT, ZH, and KO, whereas the English-derived direction was stronger for KK. None of these differences was statistically reliable, however, as all four bootstrap confidence intervals for the comparison included zero.

Gemma4-12B-IT showed a clearer and different pattern. At L26, the English-derived direction produced stronger reductions than the corresponding native direction in all four target languages. The differences were statistically reliable for IT, ZH, KO, and KK, with the largest gap observed for Kazakh. Thus, extracting a direction directly from target-language data did not improve steering performance in this model and instead produced a weaker intervention at the selected layer.

The native-vector comparison also shows that high geometric similarity does not imply behavioural equivalence. One possible explanation is that small language-dependent differences in the estimation of the behavioural direction are amplified during intervention, even when the resulting directions remain highly similar geometrically. Differences in model training exposure or translation quality may also contribute to this estimation gap. However, the present experiments cannot distinguish between these possibilities.

More generally, the native-vector comparison suggests that multilingual steering should not assume that target-language extraction is preferable by default. Whether a transferred or language-specific direction is more effective appears to depend on the model and intervention setting. Direct behavioural comparison is therefore needed when selecting steering directions for multilingual use.

5.6 LIMITATIONS AND FUTURE WORK

Several limitations constrain the conclusions that can be drawn from these experiments. First, the cross-task evaluation covers only sycophancy and TruthfulQA forced-choice truthfulness. The task-dependent reversal observed between Qwen3.5-9B and OLMo3-7B-Instruct shows that transfer patterns can differ across behaviours within the same model, but two tasks are not sufficient to characterise the full range of this dependence. Evaluating the same steering framework on additional alignment-relevant behaviours, such as hallucination, refusal, or reasoning, would help determine whether the observed dissociation reflects a stable model-level property or a task-specific interaction.

Second, the experiments cover three model families and four non-English target languages. The inclusion of Italian, Chinese, Korean, and Kazakh provides variation in language family and script, but is not sufficient to isolate which linguistic or model-level factors account for differences in steering effectiveness. A broader study across additional models and languages would be required to test whether patterns such as the stable L25–27 intervention region in Gemma4-12B-IT generalise beyond the models examined here.

Third, steering was evaluated using fixed intervention multipliers and predefined layer ranges. This design allows direct comparison across languages, but it does not determine whether each language would favour a different intervention strength or whether stronger effects exist outside the evaluated ranges. Joint sweeps over layer and steering strength could separate differences in optimal intervention depth from differences in sensitivity to the intervention magnitude.

Fourth, the geometry-behaviour analysis is based on a small number of layers and target languages. The reported correlations are therefore descriptive rather than inferential. Although the present results suggest that cosine similarity and vector norm do not consistently track behavioural effectiveness, larger layer ranges and additional models would be needed to characterise this relationship more precisely.

Finally, the norm-matched controls behaved differently across the two model families. Qwen3.5-9B retained sufficient parseability for direct behavioural comparison, whereas most randomised interventions in Gemma4-12B-IT disrupted parseable generation. The Gemma controls therefore indicate that arbitrary norm-matched directions interact with the model very differently from the learned CAA direction, but they do not provide the same behavioural baseline available for Qwen3.5-9B. Future work could use smaller perturbation magnitudes or alternative control constructions to obtain more directly comparable behavioural controls.

Combining these extensions with mechanistic analysis of where cross-lingual divergence arises would allow the gap between representational similarity and behavioural transfer identified here to be examined at a finer level. The present experiments identify cases in which highly similar multilingual representations coexist with different behavioural outcomes, but they do not determine where this divergence arises. Analysing whether the difference is introduced during prompt interpretation, intermediate reasoning, or response generation could help explain why a shared steering direction can remain geometrically aligned across languages while producing language- and model-dependent behavioural effects.

CONCLUSION

This dissertation investigated whether activation-steering directions constructed from English data transfer reliably across languages, whether such transfer generalises across behaviours, and whether cross-lingual representational similarity tracks variation in behavioural steering effects. Using Contrastive Activation Addition, we evaluated the cross-lingual effects of English-derived steering directions on sycophancy across English, Italian, Chinese, Korean, and Kazakh in three open-weight models, alongside language-specific vectors, norm-matched controls, representational geometry analyses, and a cross-task evaluation on TruthfulQA.

English-derived sycophancy steering reduced sycophancy across all four non-English target languages in Qwen3.5-9B and Gemma4-12B-IT, but did not transfer reliably in OLMo3-7B-Instruct, where only the English effect was statistically reliable. Transfer was not behaviourally uniform even where it succeeded: Qwen3.5-9B showed greater cross-lingual variation in both steering magnitude and the layer of maximum effect, whereas Gemma4-12B-IT showed more consistent layer localisation, with all five languages reaching their strongest reductions within Layers 25–27. Language-specific vector extraction did not provide a consistent advantage across models.

The cross-task evaluation revealed a behaviour-dependent dissociation. Qwen3.5-9B transferred sycophancy steering reliably but showed no reliable cross-lingual truthfulness transfer at the English-selected layer. OLMo3-7B-Instruct showed the opposite pattern: no reliable cross-lingual sycophancy transfer, but reliable positive truthfulness effects in three of four non-English

languages. A model’s cross-lingual transfer profile therefore cannot be characterised from a single behaviour.

At the representational level, language-specific steering directions were highly aligned in Qwen3.5-9B and Gemma4-12B-IT, yet neither cosine similarity nor vector norm consistently tracked behavioural steering magnitude or optimal intervention depth. OLMo3-7B-Instruct showed markedly lower cross-lingual alignment, consistent with its failure to transfer sycophancy, but this single case is not sufficient to establish that low similarity precludes transfer. High representational similarity is therefore not sufficient to guarantee comparable behavioural transfer, and whether it is necessary remains unresolved. Cross-lingual alignment evaluation should therefore include direct behavioural evidence from each target language rather than relying on geometric proxies alone.

Part I

APPENDIX

ADDITIONAL EXPERIMENTAL DETAILS

A.1 PAIRED PARSEABLE SAMPLE SIZES

Tables 12–14 report the number of paired parseable examples (N) for each model, language, and intervention layer. Values below 200 reflect cases in which either the baseline or steered output was unparseable and the example was excluded from paired behavioural calculations.

Table 12: Paired parseable sample sizes for Qwen3.5-9B ($\alpha = -1.5$, Layers 13–18).

Language	L13	L14	L15	L16	L17	L18
English (EN)	159	181	162	149	155	162
Italian (IT)	155	172	147	134	141	144
Chinese (ZH)	163	173	169	153	155	171
Korean (KO)	171	190	188	162	147	162
Kazakh (KK)	159	183	140	145	156	172

Table 13: Paired parseable sample sizes for Gemma4-12B-IT ($\alpha = -1.0$, Layers 24–28).

Language	L24	L25	L26	L27	L28
English (EN)	159	144	135	151	154
Italian (IT)	190	186	183	191	185
Chinese (ZH)	199	193	198	198	194
Korean (KO)	200	200	197	200	200
Kazakh (KK)	199	200	199	198	200

A.2 TRUTHFULQA FULL RESULTS

Table 15 reports per-language best-layer TruthfulQA results alongside the fixed English-selected layer results reported in Section 4.4. Per-language best-layer results are not used in the main analysis and are provided for reference only.

Table 14: Paired parseable sample sizes for OLMo3-7B-Instruct ($\alpha = -1.5$, Layers 12–16).

Language	L12	L13	L14	L15	L16
English (EN)	192	191	190	189	192
Italian (IT)	200	200	200	200	200
Chinese (ZH)	200	200	200	200	200
Korean (KO)	200	200	200	200	200
Kazakh (KK)	200	200	200	200	200

Table 15: TruthfulQA results for Qwen3.5-9B and OLMo3-7B-Instruct. Δ TR (pp) at the English-selected reference layer (fixed) and at the per-language best layer (oracle). Gemma4-12B-IT results are omitted due to ceiling effects at baseline (89–91%).

Language	Qwen3.5-9B		OLMo3-7B-Instruct	
	Fixed (L18)	Best Layer	Fixed (L14)	Best Layer
English (EN)	+3.0	+3.0 (L18)	+6.0	+6.0 (L14)
Italian (IT)	+0.0	+1.5 (L18)	+9.0	+9.0 (L14)
Chinese (ZH)	+1.0	+5.0 (L18)	+9.0	+9.0 (L14)
Korean (KO)	−1.0	+3.1 (L19)	+4.0	+9.0 (L12)
Kazakh (KK)	−1.0	+8.9 (L13)	+12.0	+12.0 (L14)

A.3 TRANSLATION PROMPT

The following structured prompt was used to generate all target-language translations via GPT-4o.

Translate the following JSON entries into {target_language}. Translate only the question, answer_matching_behavior, and answer_not_matching_behavior fields, along with any persona or biographical information contained in the prompt text. Preserve all formatting, line breaks, special characters, and the answer-choice labels (A) and (B) exactly as they appear in the source. The answer_matching_behavior and answer_not_matching_behavior fields must remain valid, non-empty, and distinct after translation. Return valid JSON only, with no additional commentary or markdown formatting.

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DECLARATION

I declare that this dissertation is my own work and that all sources used have been appropriately acknowledged and referenced. This work has not been submitted, in whole or in substantial part, for any other degree or qualification.

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